Filter Wizard

Sallen-Key filter

Created on 03/31/2026

ANALOG DEVICES

Filter Wizard Design Report

Filter Requirements for Band-Pass, 8th order Butterworth

Specifications: Optimize: Noise; +Vs: 5; -Vs: -5

Gain: 0 dB

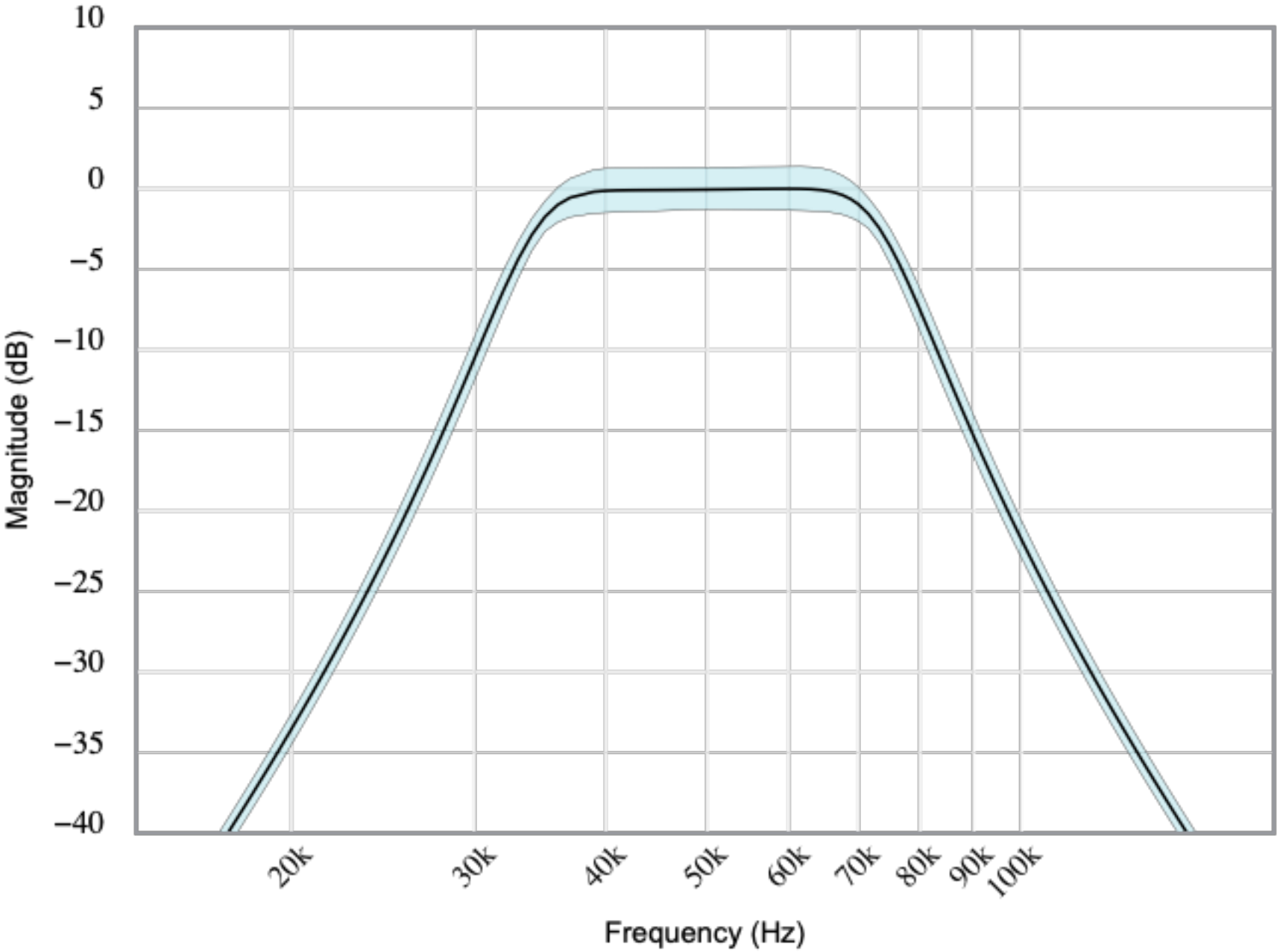
Passband: -3dB at 40kHz

Stopband: -20dB at 80kHz

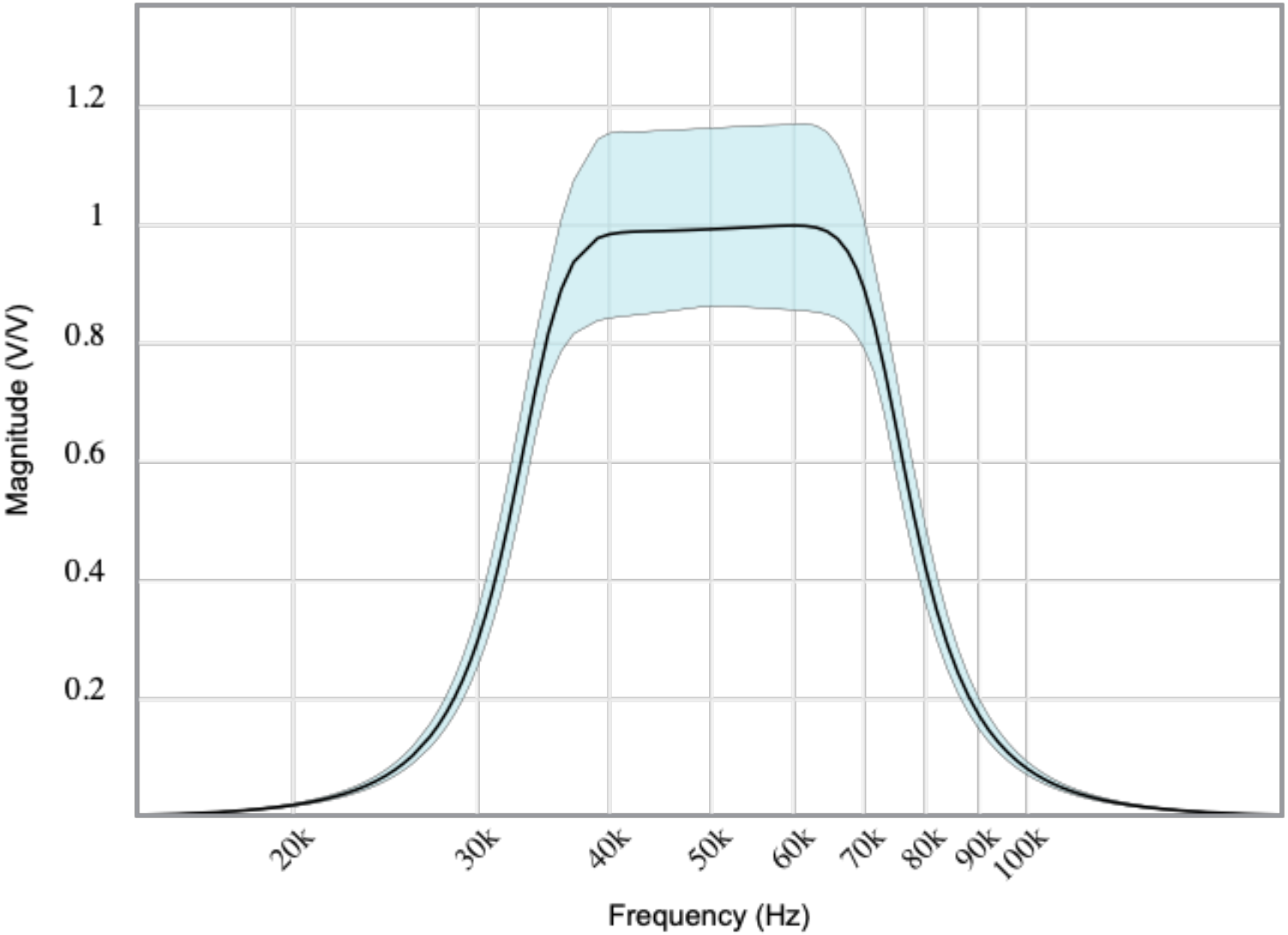
Component Tolerances: Capacitor = 1%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

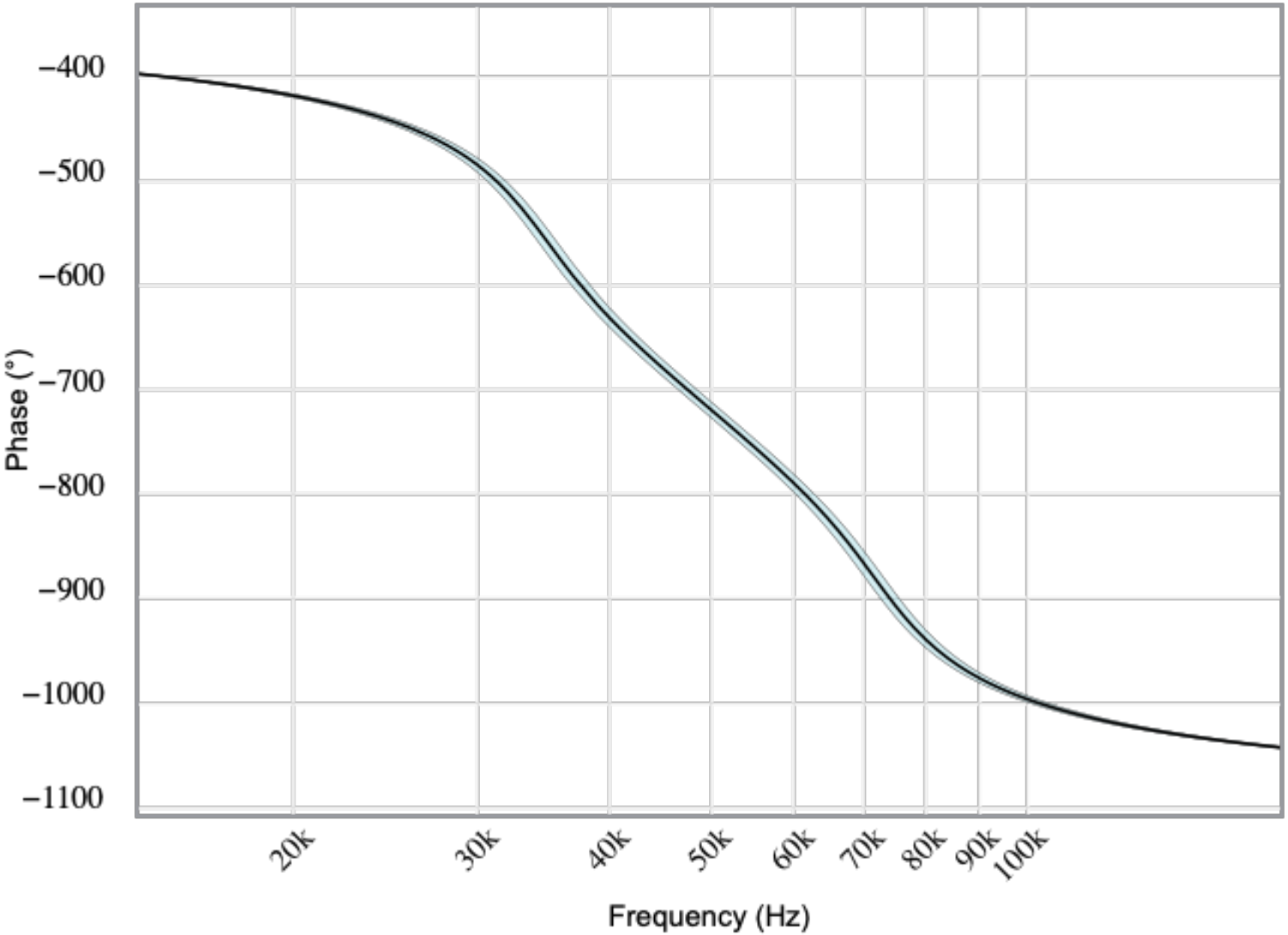
Magnitude(dB)



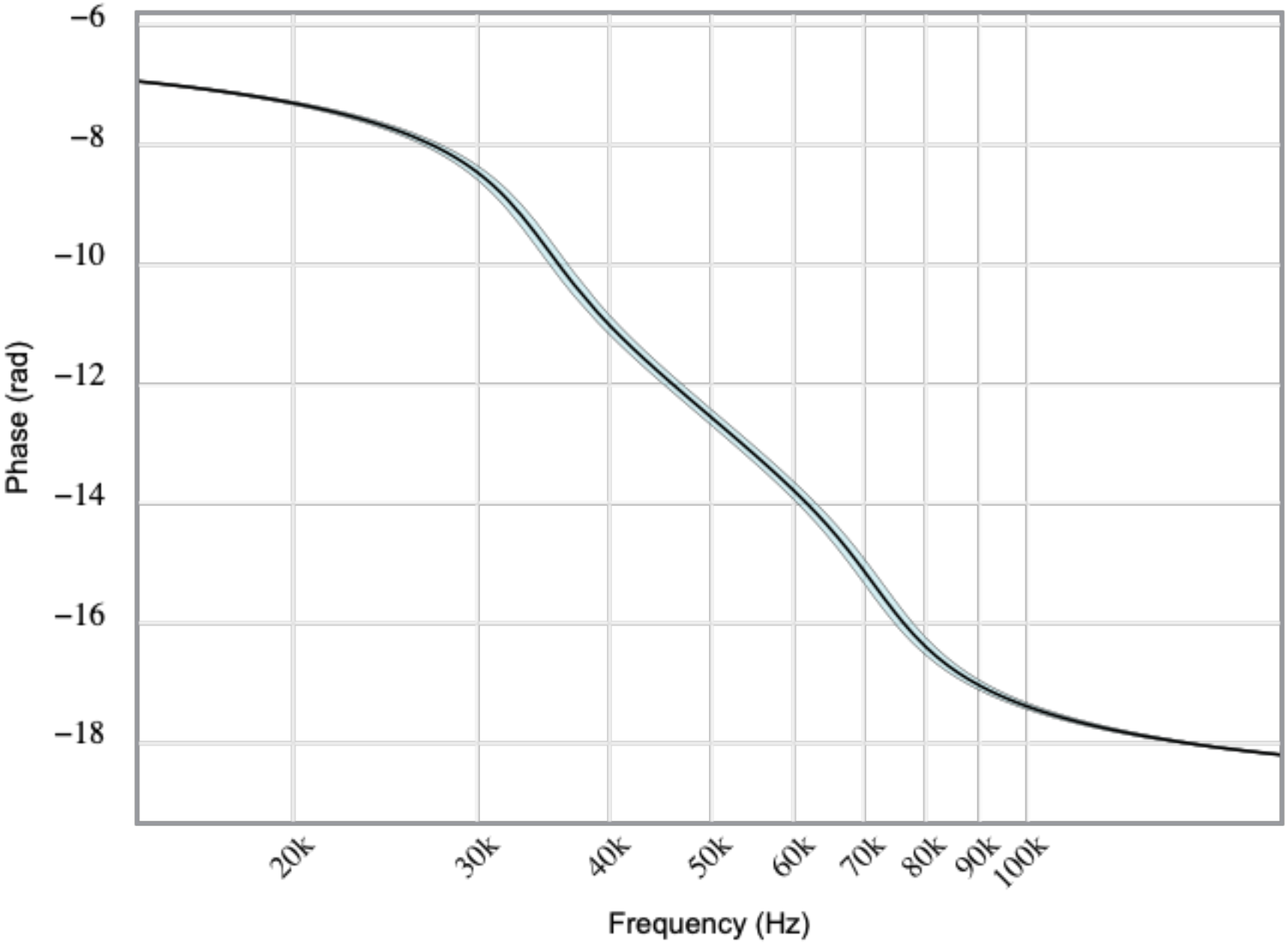
Magnitude(Volts per Volt)



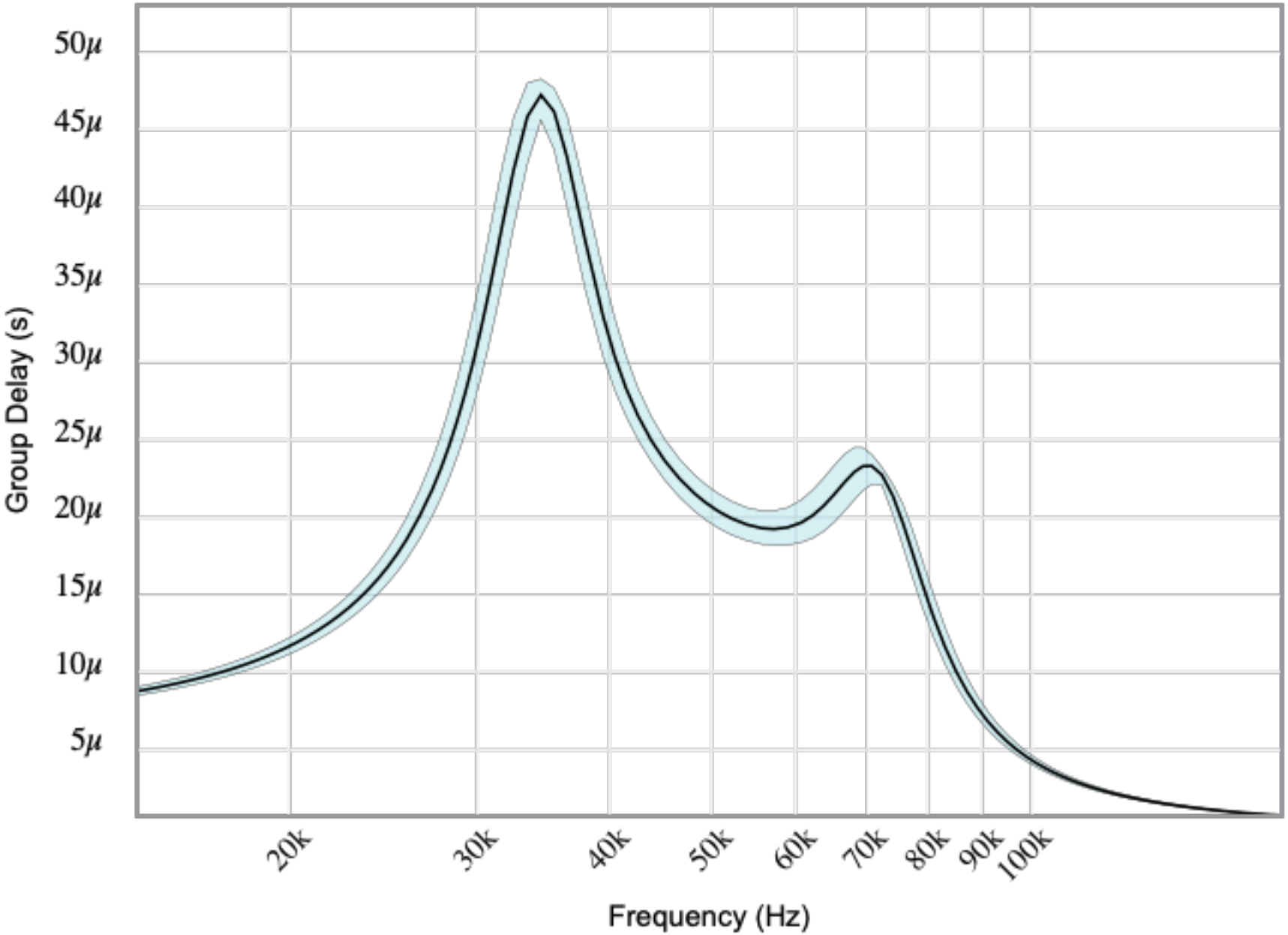
Phase(degrees)



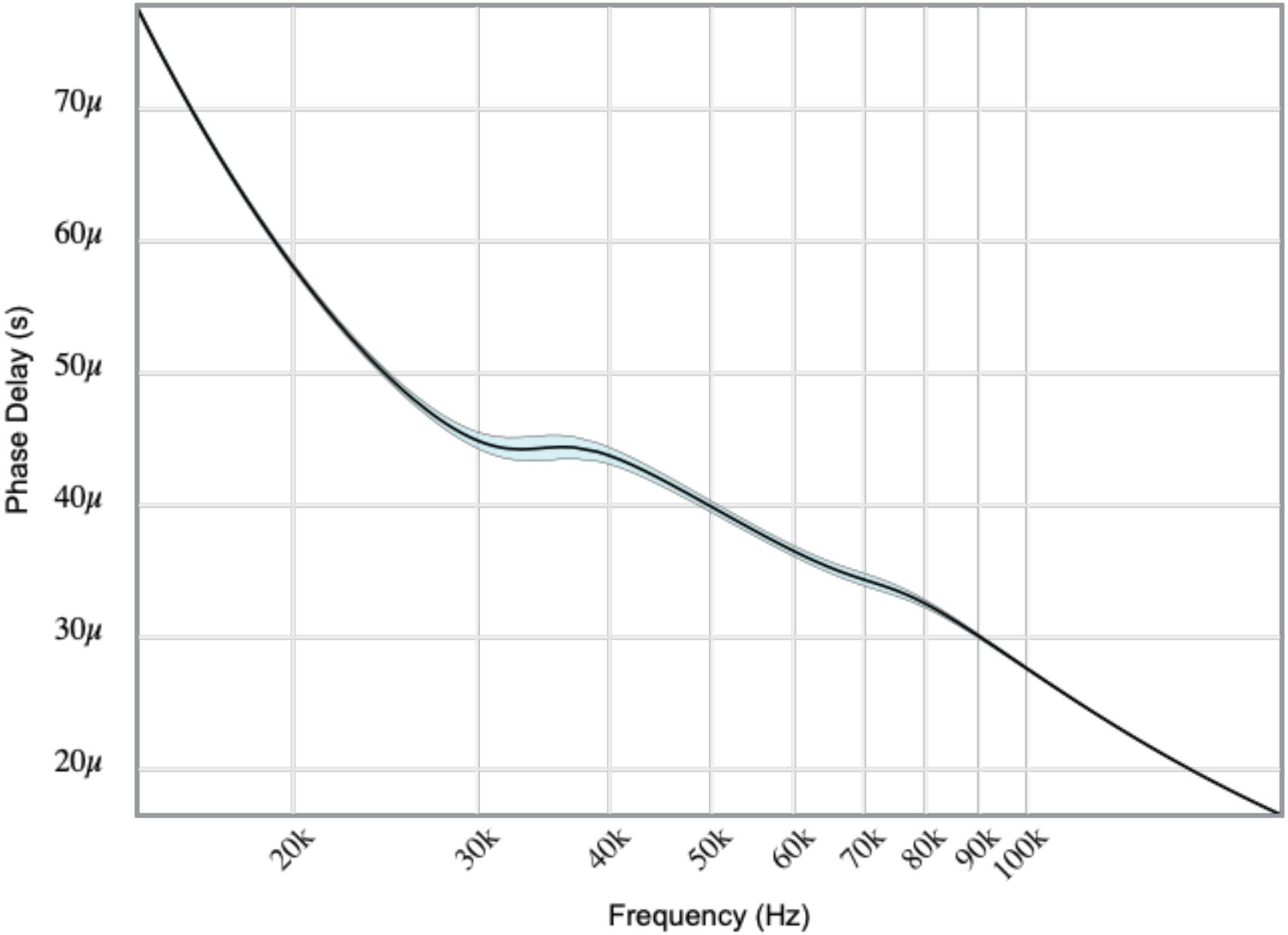
Phase(radians)



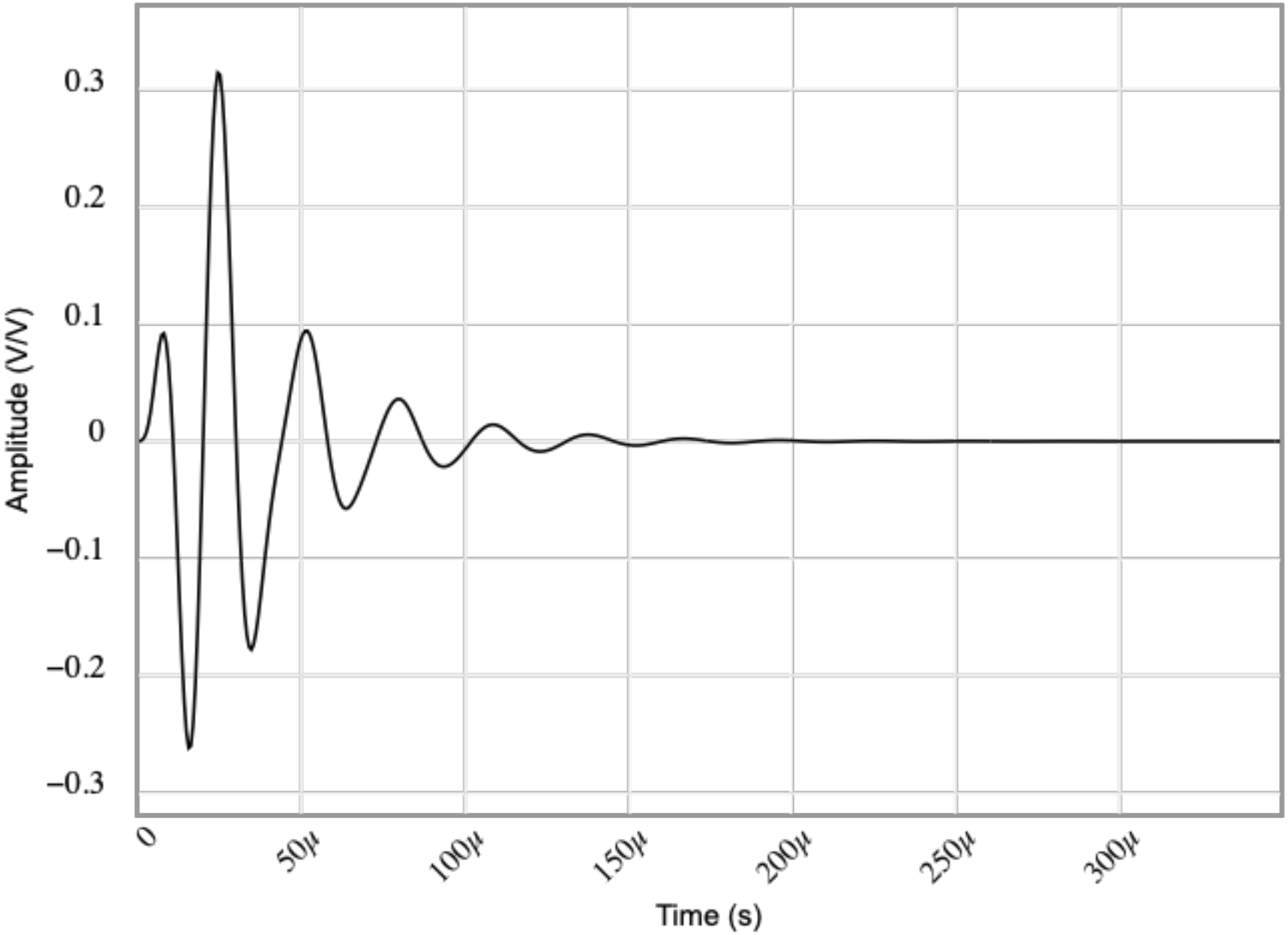
Group Delay


50μ
45μ
40μ
35μ
30μ
25μ
20μ
15μ
10μ
5μ
Group Delay (s)
20k
30k
40k
50k
60k
70k
80k
90k
100k
Frequency (Hz)

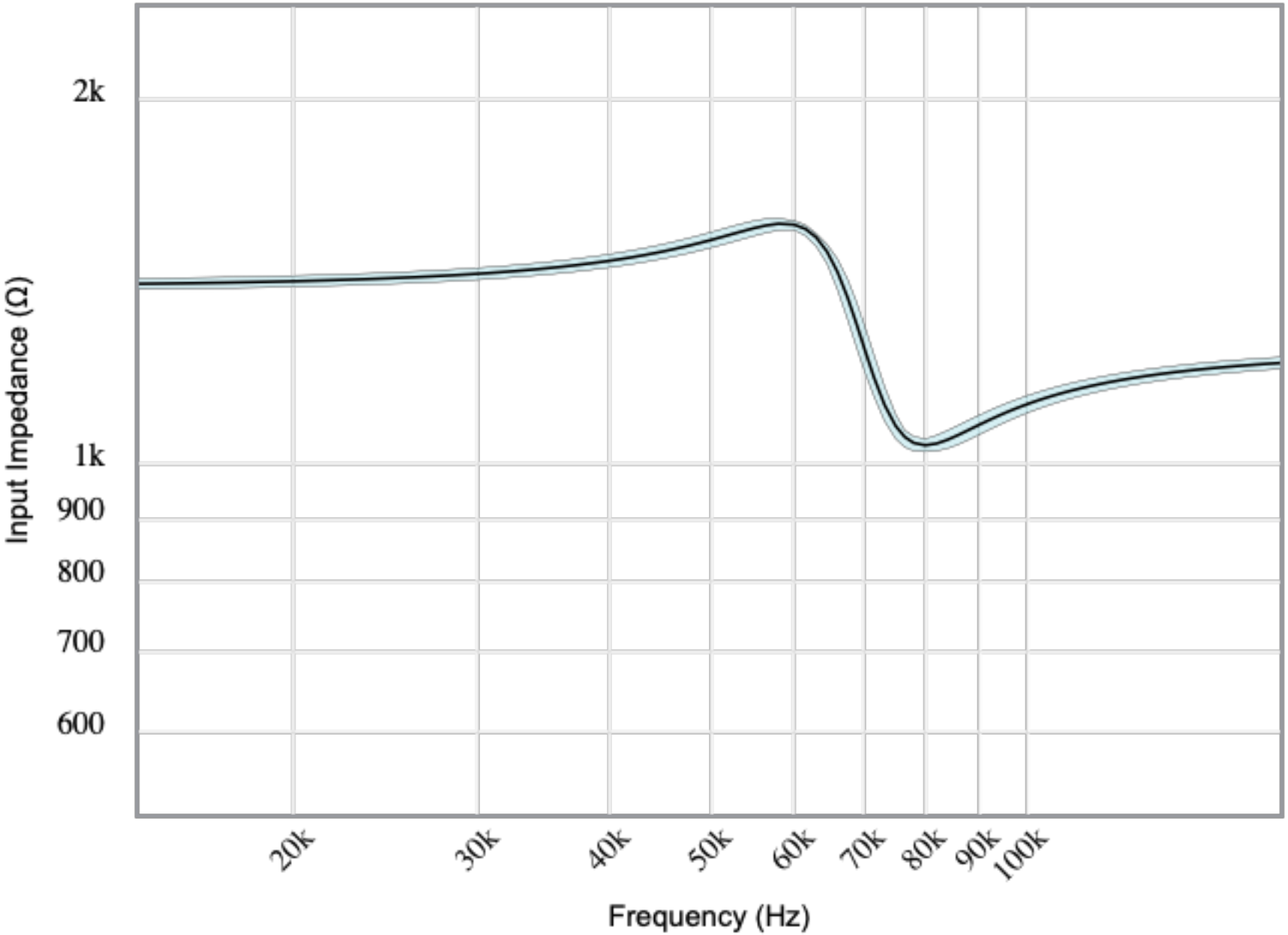
Phase Delay


70μ
60μ
50μ
40μ
30μ
20μ
Phase Delay (s)
20k
30k
40k
50k
60k
70k
80k
90k
100k
Frequency (Hz)

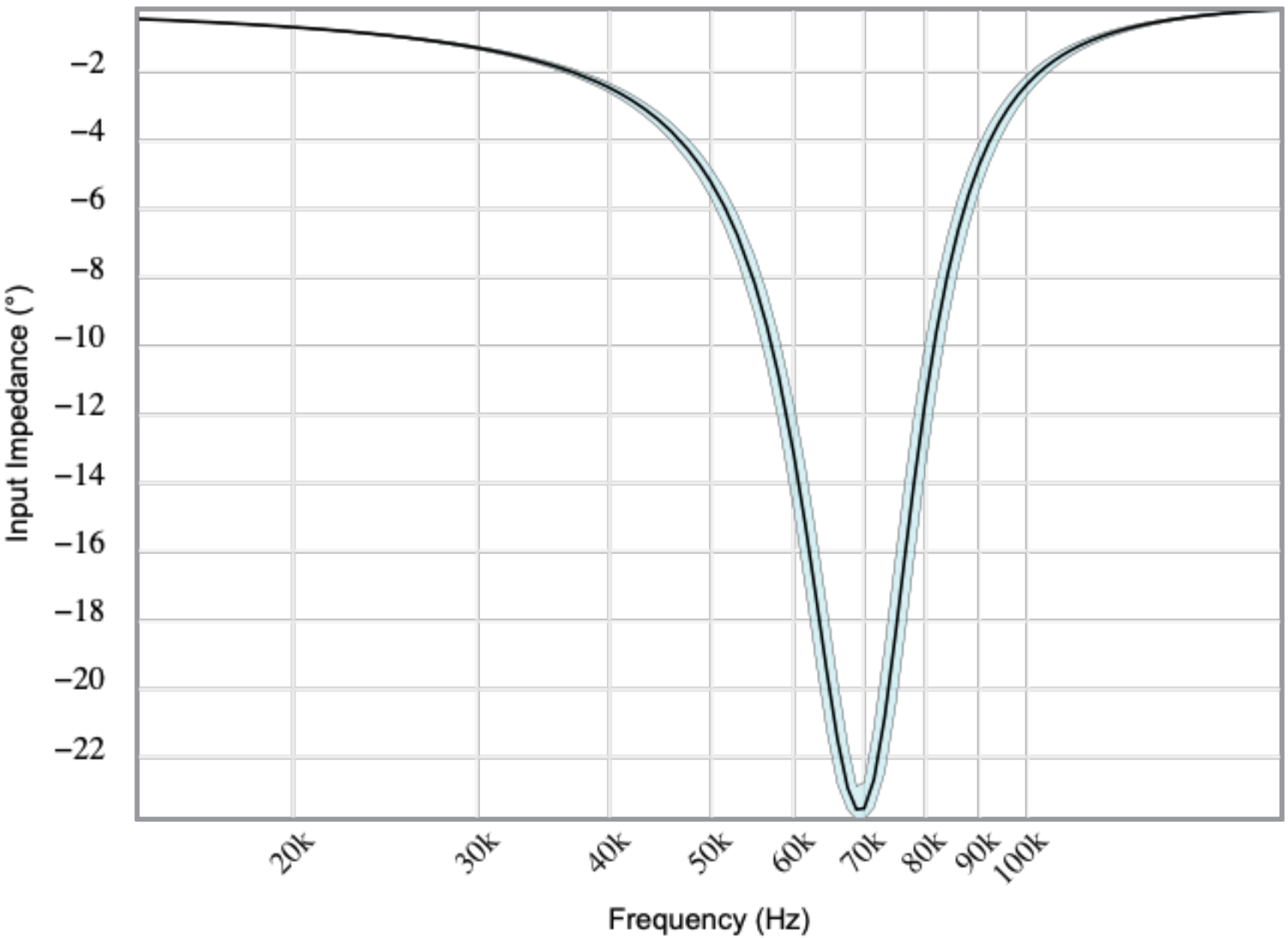
Step Response



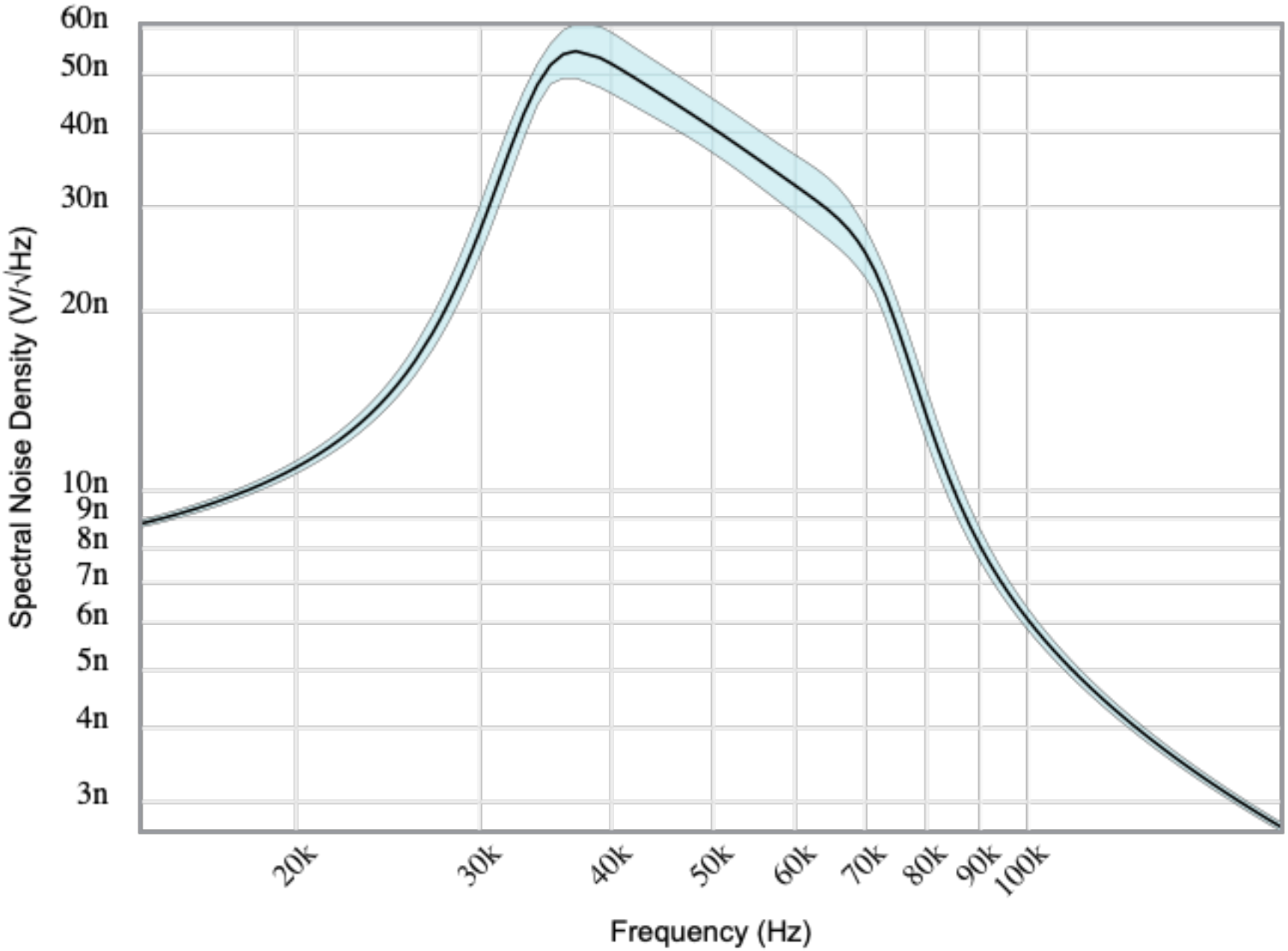
Input Impedance Magnitude



Input Impedance Phase



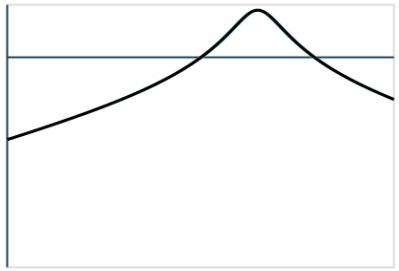
Noise

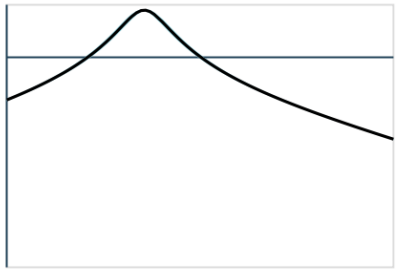

60n
50n
40n
30n
20n
10n
9n
8n
7n
6n
5n
4n
3n
Spectral Noise Density (V/√Hz)
20k
30k
40k
50k
60k
70k
80k
90k
100k
Frequency (Hz)

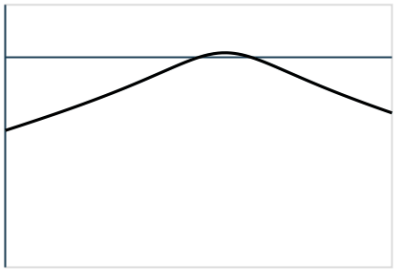
Stages

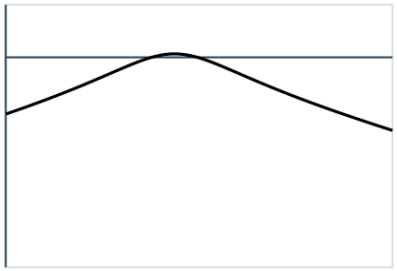
Your filter requires 4 op amp stage(s) with the following characteristics

	A		B		C		D	
	2nd order Band-Pass Multiple Feedback		2nd order Band-Pass Multiple Feedback		2nd order Band-Pass Multiple Feedback		2nd order Band-Pass Multiple Feedback	
	Target	**Simulated**	**Target**	**Simulated**	**Target**	**Simulated**	**Target**	**Simulated**
Gain (V/V):	2.79	2.76 to 2.87	2.79	2.76 to 2.87	1.1	1.08 to 1.13	1.1	1.06 to 1.1
f_p (Hz):	72.1k	70.8k to 73.7k	34.7k	34k to 35.4k	58.9k	58k to 60.4k	42.5k	41.6k to 43.3k
Q:	3.48	3.46 to 3.53	3.48	3.46 to 3.53	1.37	1.36 to 1.39	1.37	1.35 to 1.37

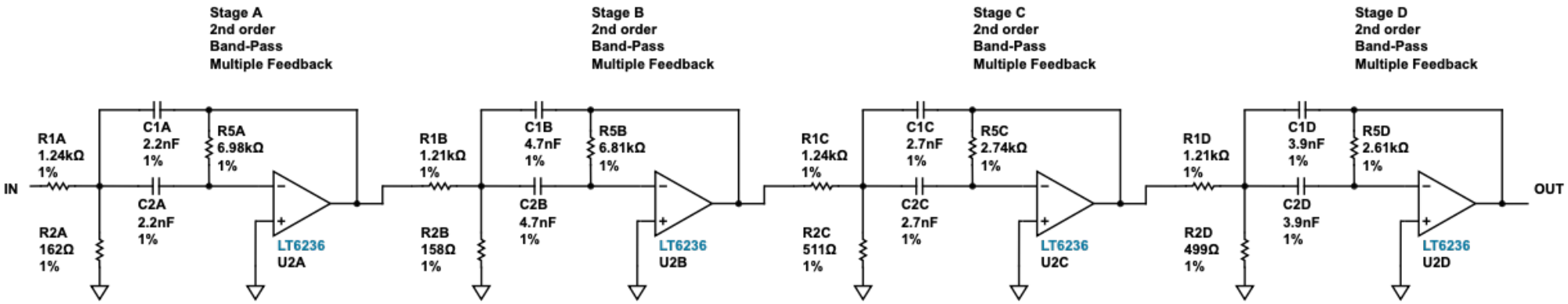
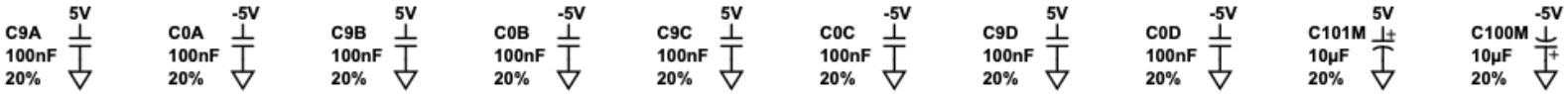








Circuit



BYPASS CAPACITORS